

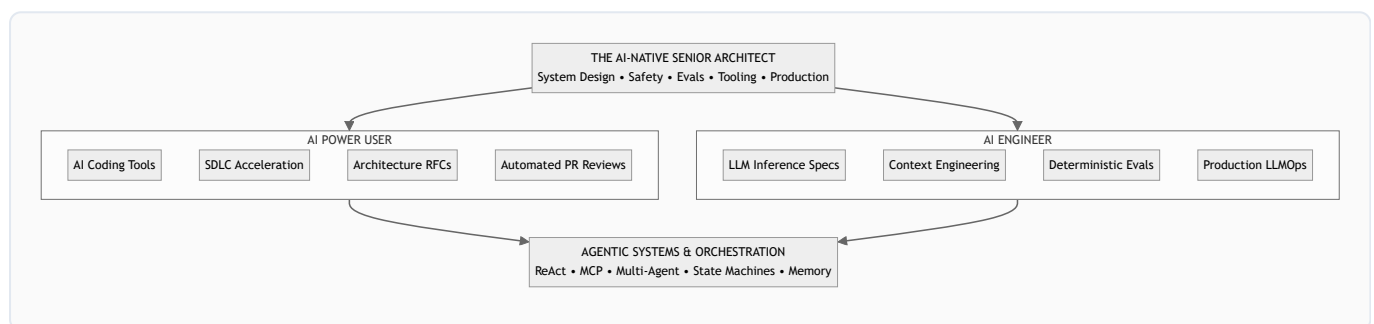
Senior Engineer to AI Engineer Roadmap

Live GitHub Repository & Documentation Hub

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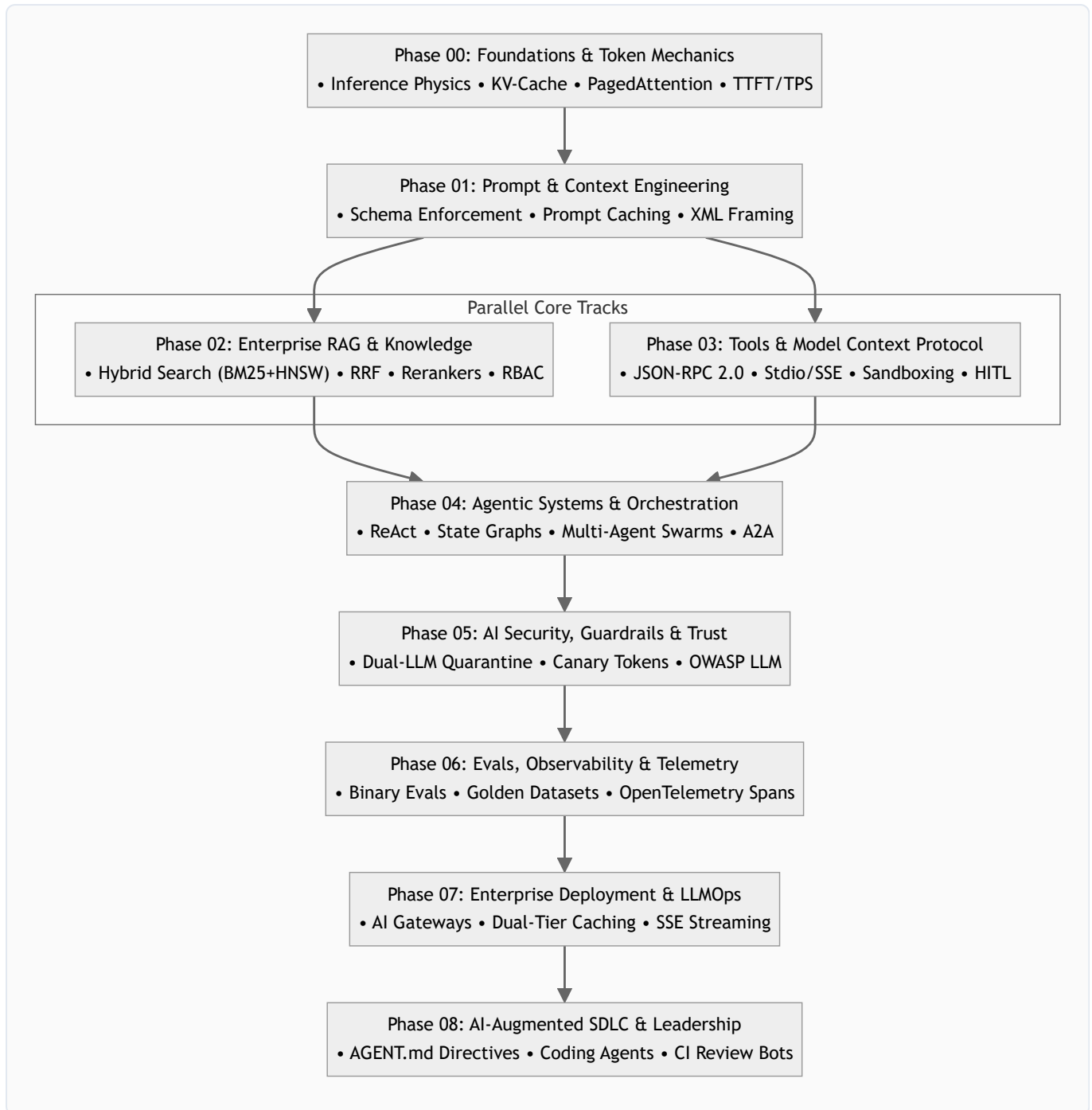
AI Engineer & Agentic Systems Roadmap: Senior & Lead Developer Edition

A production-focused engineering curriculum and architectural reference designed for Senior Engineers, Tech Leads, and Software Architects building enterprise AI applications and autonomous agentic systems.



Master Curriculum Progression

The curriculum is structured sequentially from inference physics to autonomous multi-agent systems and engineering leadership:



Curriculum Structure & Phased Syllabus

Phase	Directory	Focus & Key Deliverables	Estimated Time	Level
00	Foundations & Token Mechanics	Test-time compute & reasoning tokens (o1, o3-mini, Claude 3.7, DeepSeek R1), KV-Cache VRAM sizing, TTFT vs TPS, PagedAttention.	1-2 Weeks	Core
01	Prompt & Context Engineering	Prompt hierarchy, Anthropic XML tags, FSM constrained grammar decoding, 5-min ephemeral prompt caching economics.	2 Weeks	Core
02	RAG & Enterprise Knowledge	Layout-aware parsing, Hybrid Search (Dense HNSW + BM25), Reciprocal Rank Fusion (RRF), Cross-Encoder rerankers, RBAC.	2-3 Weeks	Advanced
03	Tools & Model Context Protocol	MCP JSON-RPC 2.0 wire protocol, FastMCP servers, Agent Skills, schema validation, container sandboxing, HITL approval.	2 Weeks	Advanced
04	Agentic Systems & Orchestration	Modern frameworks (PydanticAI, OpenAI Agents SDK, LangGraph, Google ADK, Semantic Kernel), ReAct loops, state checkpointing, A2A swarms.	3 Weeks	Architect
05	AI Security, Guardrails & Trust	OWASP LLM Top 10, Dual-LLM Privilege Separation (Quarantine), canary tokens, NeMo / Llama Guard 3, ephemeral sandboxes.	1-2 Weeks	Architect
06	Evals, Observability & Telemetry	Hamel Husain 3-level evaluation model, binary LLM-as-a-judge rubrics, golden test sets, OpenTelemetry GenAI spans.	2 Weeks	Architect
07	Production Deployment & LLMOps	Serverless vs vLLM, LiteLLM Resilient AI Gateway, dual-tier caching (Exact SHA-256 + Vector), SSE streaming.	2 Weeks	Lead/Ops
08	AI-Augmented SDLC & Leadership	Autonomous coding agents (Claude Code, Cursor, Windsurf), <code>AGENT.md</code> contracts, AI-driven TDD verification, automated PR review.	Ongoing	Executive
Playbook	Senior Transition Guide	Software 1.0 \$to\$ 3.0 shift, 6 enterprise use cases, 6 hands-on practice labs, and 90-day execution roadmap.	1 Week	Staff/Lead
Interview	80/20 Interview Prep Sheet	Master System Design Blueprints, 25 architect technical Q&As, tradeoff cheat matrices, and hardware formulas.	Continuous	Master
Resource Map	Comprehensive Resource Map	Phase-by-phase reference linking all 24 topics to official documentation, courses, and GitHub repositories.	Reference	All Levels

Architectural Mastery Tiers

Every topic across this roadmap is tagged with a 3-tier classification to focus engineering energy where it delivers maximum ROI:

Tier	Meaning & Scope	Focus Allocation
[MUST-HAVE] 	Production Invariants & Core Architecture: Essential for enterprise applications, production reliability, immediate business ROI, and system architecture. Non-negotiable foundation.	80% Focus
[GOOD-TO-HAVE] 	Advanced Scaling & Complex Orchestration: Swarm handoffs, specialized memory graphs, custom guardrails, latency optimizations, and high-concurrency scaling.	15% Focus
[KNOWLEDGE-BASE] 	Conceptual Reference & Architectural Intuition: Mathematical derivations, training physics, and hardware formulas. Understand mental models; skip coding from scratch.	5% Focus

 Recommended Reading Paths

Track	Objective	Target Phases	Key Outcome
Path A	Enterprise Knowledge & RAG	Phases 00 \$to\$ 01 \$to\$ 02 \$to\$ 06	Production grounded search with hybrid retrieval, cross-encoder reranking, and discrete binary evaluation.
Path B	Autonomous Tool-Using Agents	Phases 01 \$to\$ 03 \$to\$ 04 \$to\$ 05	Stateful agents with strict JSON schemas, MCP tool integration, durable state machines, and dual-LLM quarantine.
Path C	LLMOps, Infrastructure & Leadership	Phases 07 \$to\$ 08 \$to\$ Playbook \$to\$ Interview	Resilient multi-provider gateways, OpenTelemetry tracing, <code>AGENT.md</code> repository directives, and system design mastery.

 Hands-On Practice Labs

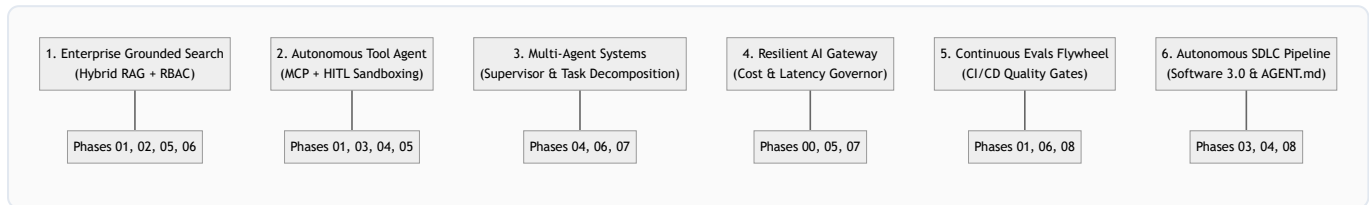
Master production patterns by building and verifying these standalone reference implementations:

Lab	Name	Core Architectural Deliverable	Lab File
01	Multi-Tenant Hybrid RAG	BM25 + Dense HNSW search, Reciprocal Rank Fusion, and cross-encoder reranking with tenant isolation.	View Lab
02	Tool Execution with MCP	Model Context Protocol JSON-RPC 2.0 server & client with schema validation and container sandboxing.	View Lab
03	Stateful Agent Orchestration	Directed state machine with graph reducers, durable SQLite checkpointing, and execution pause/resume.	View Lab
04	Agent Failure Defense	Rolling SHA-256 action loop detection, blast radius previews, and automated context compaction at 75% capacity.	View Lab
05	AI Observability & Tracing	Distributed tracing with OpenTelemetry GenAI semantic spans, token metrics, and Jaeger/Langfuse export.	View Lab
06	Dual-LLM Quarantine & Guardrails	Unprivileged Reader LLM parsing untrusted input, cryptographic canary tokens, and egress leakage filters.	View Lab

For comprehensive module-specific capstones (e.g., *Code Review Engine*, *CI/CD Eval Gate*, *Resilient AI Gateway*), see each module's `Labs/` directory.

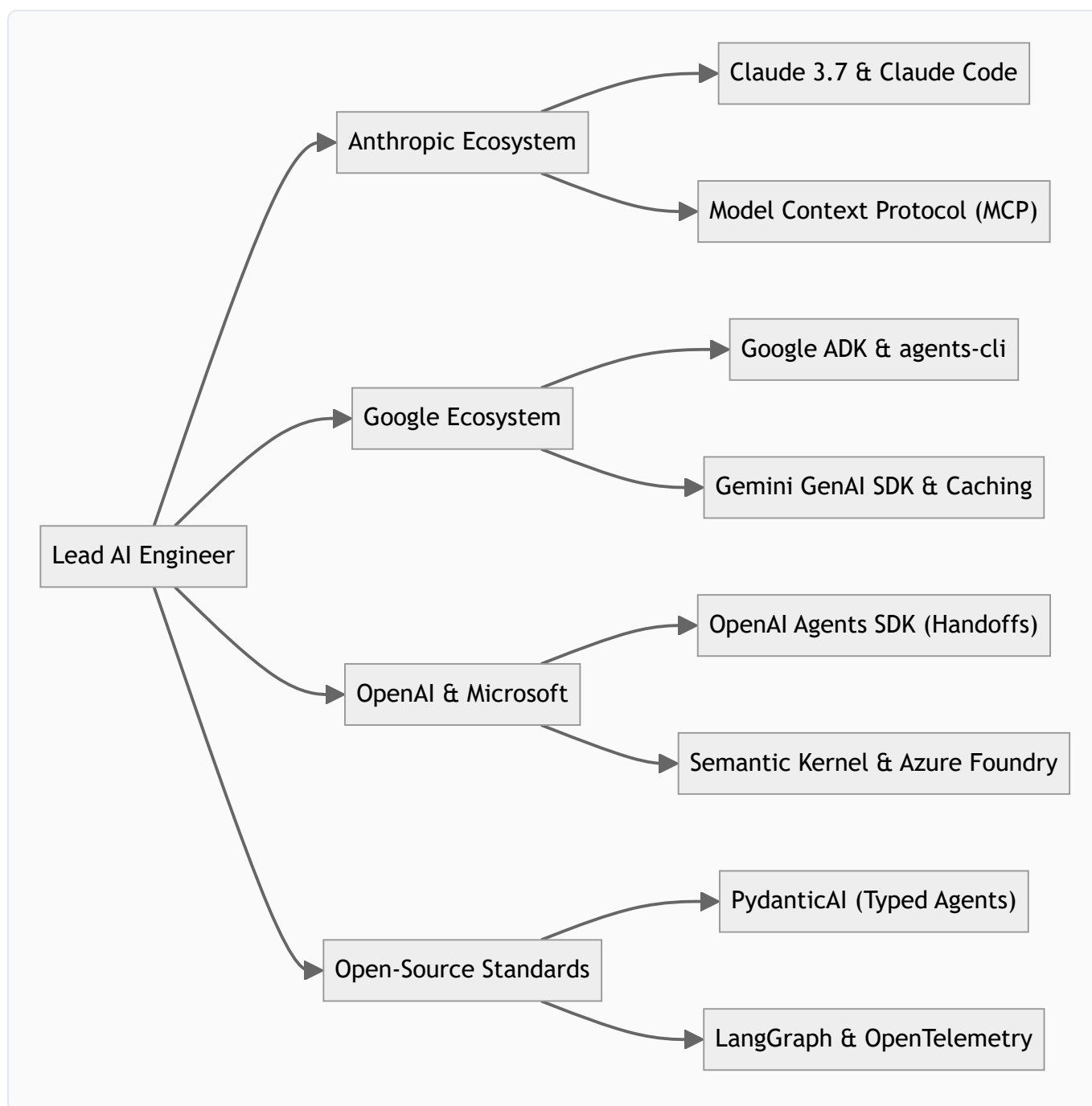
Enterprise Architecture Blueprints

The curriculum maps directly to the six primary enterprise AI architectural archetypes:



Detailed architectural specifications, failure modes, and code samples for each blueprint are in `senior-transition-guide.md`.

Ecosystem Alignment



Quick Navigation & Reference Hub

-  **The Senior Transition Guide:** The Software 1.0 to 3.0 shift, polyglot matrix, and 90-day execution plan.
-  **80/20 System Design Interview Prep:** 5 master blueprints, 25 architect Q&As, and tradeoff cheat sheets.
-  **Comprehensive Resource Map:** Direct links to official provider docs, SDKs, and courses across 24 phases.
-  **Master Resource Index:** Curated documentation, seminal papers, and enterprise frameworks.